

Assignment 4

Name: _____

Student ID: _____

12 problems. Show all of your work.

Problem 1.

- A. Find an equation of the set of all points equidistant from the points $A(-1, 5, 3)$ and $B(6, 2, -2)$.
- B. Describe the geometrical shape corresponding to the set.

Problem 2.

Describe the following regions and graphs.

- A. $x^2 + y^2 + z^2 \leq 4, z \geq 0$
- B. $x^2 + y^2 + z^2 \leq 4, z = 0$
- C. $x^2 + y^2 + z^2 = 4, z \geq 0$
- D. $x^2 + y^2 + z^2 = 4, z = 0$

Problem 3.Consider the point $(3, 7, -5)$.

- (a) What is its distance to the yz -plane?
- (b) What is its distance to the y -axis?

Problem 4. *A Solid Bounded by a Paraboloid, a Plane, and a Cylinder*

Consider the solid region bounded below by the paraboloid $z = 1 - x^2 - y^2$, above by the plane $z = 1$, and on the outside by the cylinder $x^2 + y^2 = 1$. Describe this solid.

Problem 5.

Write each combination of vectors as a single vector. See Figure 1.

- A. $\vec{PQ} + \vec{QR}$
- B. $\vec{QS} - \vec{PS}$
- C. $\vec{RS} + \vec{SP} + \vec{PQ}$

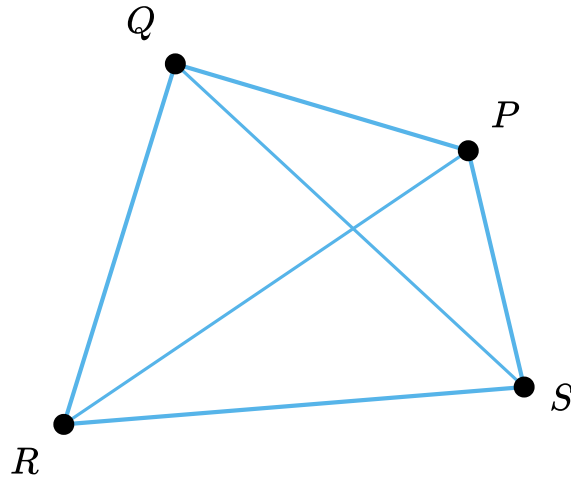


Figure 1. The quadrilateral with vertices P , Q , R , and S , with its diagonals.

Problem 6.

Find a vector of length 3 pointing in the opposite direction from the vector $2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$.

Problem 7.

Let $\vec{v}$ be the vector that goes from the point $(5, -4, \sqrt{3})$ to the point $(3, -1, 0)$. Find a vector of length $\frac{1}{3}$ that points in the same direction as $\vec{v}$.

Problem 8. *A parallelogram in space*

Consider the four points

$$A(1, 2, 5), \quad B(4, 3, 3), \quad C(6, 7, 4), \quad D(3, 6, 6).$$

- A. Show that $ABCD$ is a parallelogram.
- B. Find the point M at which the two diagonals of $ABCD$ meet.
- C. Find the perimeter of $ABCD$.

Problem 9.

Consider the two vectors $\mathbf{u} = 3\mathbf{j} + 4\mathbf{k}$ and $\mathbf{v} = \mathbf{i} + \mathbf{j}$. Write $\mathbf{u}$ as the sum of a vector parallel to $\mathbf{v}$ and a vector orthogonal to $\mathbf{v}$.

Problem 10.

Determine whether the following pairs of vectors are orthogonal, parallel, or neither.

- A. $\vec{u}_1 = -3\mathbf{i} + 9\mathbf{j} + 6\mathbf{k}$ and $\vec{u}_2 = 4\mathbf{i} - 12\mathbf{j} - 8\mathbf{k}$
- B. $\vec{v}_1 = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\vec{v}_2 = \mathbf{i} + \mathbf{j} + \mathbf{k}$
- C. $\vec{w}_1 = a\mathbf{i} + b\mathbf{j}$ and $\vec{w}_2 = c\mathbf{k}$, where a , b , and c are non-zero real numbers.

Problem 11.

A 100-meter dash is run on a track in the direction of the vector $2\mathbf{i} + 6\mathbf{j}$. The wind velocity is $\mathbf{v} = 5\mathbf{i} + 2\mathbf{j}$ km/hr. The rules say that a legal wind speed measured in the direction of the dash must not exceed 5 km/hr.

- A. Will the race results be disqualified due to an illegal wind? Justify your answer.
- B. Find the velocity vector of the wind in the direction of the dash.

Problem 12.

A constant force is

$$\mathbf{F} = \langle 2, -1, 3 \rangle.$$

A particle is constrained to move in directions parallel to the plane

$$2x - y + 2z = 0.$$

For a displacement of one unit, the work done by the force is $W = \mathbf{F} \cdot \mathbf{d}$, where $\mathbf{d}$ is an allowed unit direction lying in the plane.

- (a) Find the unit direction in the plane that maximizes the work.
- (b) Find the maximum possible work per unit distance.
- (c) Find one unit direction in the plane for which the work is zero.